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United States Patent	12401753
Kind Code	B2
Date of Patent	August 26, 2025
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Information processing apparatus, non-transitory computer readable medium storing information processing program, and information processing method

Abstract

An information processing apparatus includes a processor configured to in a case where an instruction to apply a setting of a second image processing device to a setting of a first image processing device is received, execute an application process of applying a setting of a second virtual image processing device to a setting of a first virtual image processing device, the first virtual image processing device being a virtual device of the first image processing device and being a device having the setting synchronized with the setting of the first image processing device, and the second virtual image processing device being a virtual device of the second image processing device and being a device having the setting synchronized with the setting of the second image processing device.

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Appl. No.:	17/899597
Filed:	August 30, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230291855 A1	Sep. 14, 2023

Foreign Application Priority Data

JP	2022-038630	Mar. 11, 2022
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Publication Classification

Int. Cl.: G06K15/00 (20060101); G06K15/02 (20060101); H04N1/00 (20060101)

U.S. Cl.:

CPC H04N1/00931 (20130101); H04N1/00925 (20130101); H04N1/0097 (20130101);

Field of Classification Search

CPC: H04N (1/00931); H04N (1/00925); H04N (1/0097); H04N (2201/0094); H04N (2201/0091); H04N (2201/0093); H04N (2201/0081)

USPC: 358/1.13

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Primary Examiner: Diaby; Moustapha

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-038630 filed Mar. 11, 2022.

BACKGROUND

(i) Technical Field

(2) The present disclosure relates to an information processing apparatus, a non-transitory computer readable medium storing an information processing program, and an information processing method.

(ii) Related Art

(3) JP2012-174279A discloses a printing system having a purpose of providing a seamless access to a printer to a client regardless of a printing protocol used in each printing job.

(4) The printing system includes a printing processing apparatus and a virtual printer. The printing processing apparatus connects a client system and a physical printer to each other by an interface, receives a printing job from the client system, and functions to process the printing job for performing printing by using the physical printer. In addition, the printing processing apparatus includes a printer object that is connected to the physical printer by communication and has a printing function of the physical printer. The virtual printer detects a printing protocol associated with the printing job from the client system. In addition, the virtual printer reconfigures the printing job to process the printing job in accordance with a printing protocol of the physical printer in a case where the detected printing protocol does not coincide with a printing protocol of the physical printer.

(5) JP2020-119016A discloses a printing device for providing a printer having a high probability of acquiring print data by an appropriate protocol in a case where the print data is acquired from a cloud print service.

(6) The printing device is compatible with the cloud print service and includes a unit that acquires information regarding a printer queue associated with the printing device, a unit that acquires user authentication setting information of a printer, a unit that selects one of a plurality of job information acquisition units based on the acquired information regarding the printer queue and the user authentication setting information of the printer, and a unit that acquires a list of job information from a cloud by using the selected job acquisition unit.

SUMMARY

(7) However, the techniques disclosed in JP2012-174279A and JP2020-119016A have a problem that it is not possible to reflect the setting of another image forming device to an image forming device to be used. Such a problem may occur not only in the image forming device but also in another image processing device such as a scanner device.

(8) Aspects of non-limiting embodiments of the present disclosure relate to an information processing apparatus, a non-transitory computer readable medium storing an information processing program, and an information processing method capable of, in a configuration in which an image processing device and a virtual image processing device corresponding to the image processing device synchronize a setting, reflecting setting of the image processing device to the other image processing device to be used.

(9) Aspects of certain non-limiting embodiments of the present disclosure overcome the above disadvantages and/or other disadvantages not described above. However, aspects of the non-limiting embodiments are not required to overcome the disadvantages described above, and aspects

of the non-limiting embodiments of the present disclosure may not overcome any of the disadvantages described above.

(10) According to an aspect of the present disclosure, there is provided an information processing apparatus includes a processor configured to in a case where an instruction to apply a setting of a second image processing device to a setting of a first image processing device is received, execute an application process of applying a setting of a second virtual image processing device to a setting of a first virtual image processing device, the first virtual image processing device being a virtual device of the first image processing device and being a device having the setting synchronized with the setting of the first image processing device, and the second virtual image processing device being a virtual device of the second image processing device and being a device having the setting synchronized with the setting of the second image processing device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:
- (2) FIG. 1 is a block diagram illustrating an example of a configuration of an information processing system according to an exemplary embodiment of the present disclosure;
- (3) FIG. 2 is a block diagram illustrating an example of a hardware configuration of a server according to the exemplary embodiment of the present disclosure;
- (4) FIG. 3 is a block diagram illustrating an example of a hardware configuration of a terminal device according to the exemplary embodiment of the present disclosure;
- (5) FIG. 4 is a block diagram illustrating an example of a functional configuration of a server according to a first exemplary embodiment of the present disclosure;
- (6) FIG. 5 is a schematic diagram illustrating an example of a configuration of a virtual printer management information database according to the exemplary embodiment of the present disclosure;
- (7) FIG. 6 is a flowchart illustrating an example of a printing instruction process according to the exemplary embodiment of the present disclosure;
- (8) FIG. 7 is a front view illustrating an example of a printing instruction screen according to the exemplary embodiment of the present disclosure;
- (9) FIG. 8 is a flowchart illustrating an example of information processing according to the first exemplary embodiment of the present disclosure;
- (10) FIG. 9 is a flowchart illustrating an example of a verification process according to the first exemplary embodiment of the present disclosure;
- (11) FIG. 10 is a front view illustrating an example of a setting selection screen according to the exemplary embodiment of the present disclosure;
- (12) FIG. 11 is a block diagram illustrating an example of a functional configuration of a server according to a second exemplary embodiment of the present disclosure;
- (13) FIG. 12 is a flowchart illustrating an example of information processing according to the second exemplary embodiment of the present disclosure; and
- (14) FIG. 13 is a flowchart illustrating an example of a verification process according to the second exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

(15) Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the drawings. In the present exemplary embodiment, a case where the technique of the present disclosure is applied to an information processing system that uses, as a target, a plurality of image forming devices scattered in various places including a shared office and on-site, and

provides a cloud print service that uses each image forming device as a virtual printer will be described. The “cloud print service” referred here is represented by “UNIVERSAL PRINT”, which is a cloud-compatible print service provided by Microsoft Corporation. By using this service, it is possible to perform printing by an image forming device via a cloud even from a terminal device that is a going destination and is not allowed to be connected to a network in which the image forming device is installed.

First Exemplary Embodiment

(16) First, the configuration of an information processing system **1** according to the present exemplary embodiment will be described with reference to FIG. **1**. FIG. **1** is a block diagram illustrating an example of the configuration of the information processing system **1** according to the present exemplary embodiment.

(17) As illustrated in FIG. **1**, the information processing system **1** according to the present exemplary embodiment includes a server **10** as an information processing apparatus in the technique of the present disclosure, a plurality of terminal devices **30A**, **30B**, . . . , and a plurality of image forming devices **50A**, **50B**. In a case where the terminal devices **30A**, **30B**, . . . are described without distinguishment, the terminal devices **30A**, **30B**, . . . are simply collectively referred to as a “terminal device **30**” below. In addition, in a case where the image forming devices **50A**, **50B**, are described without distinguishment, the image forming devices **50A**, **50B**, . . . are simply collectively referred to as an “image forming device **50**” below.

(18) Examples of the server **10** and the terminal device **30** include information processing apparatuses such as a personal computer and a server computer. In the present exemplary embodiment, a digital multifunction device having an image printing function, an image scanning function, an image transmission function, and the like is applied as the image forming device **50**. The present disclosure is not limited to this form. A form in which another image forming device such as an image forming device having only an image printing function or an image forming device having only an image printing function and an image scanning function is applied as the image forming device **50** may be made.

(19) All the image forming devices **50A**, **50B**, . . . are not limited to having the identical specifications. Services that can be performed may vary, items settable in the service may vary, and equipped optional items may vary.

(20) The server **10**, the terminal device **30**, and the image forming device **50** are connected to each other via a network **N**. The server **10** is configured so that the server **10** can communicate with the terminal device **30** and the image forming device **50** via the network **N**.

(21) In the present exemplary embodiment, a combination of a public communication line such as the Internet and a telephone line network and a communication line in a company such as a local area network (LAN) and a wide area network (WAN) is used as the network **N**, but the present disclosure is not limited to this form. For example, a form in which only one of the above public communication line and the communication line in the company is applied as the network **N** may be made. Further, in the present exemplary embodiment, the wired and wireless communication lines are applied as the network **N**, but the present disclosure is not limited to this form. A form in which only one of the wireless communication line and the wired communication line is applied may be made.

(22) Next, the configuration of the server **10** according to the present exemplary embodiment will be described with reference to FIG. **2**. FIG. **2** is a block diagram illustrating an example of the hardware configuration of the server **10** according to the present exemplary embodiment.

(23) As illustrated in FIG. **2**, the server **10** according to the present exemplary embodiment includes a central processing unit (CPU) **11**, a memory **12** as a temporary storage area, a non-volatile storage unit **13**, an input unit **14** such as a keyboard and a mouse, a display unit **15** such as a liquid crystal display, a medium reading and writing (R/W) device **16**, and a communication interface (I/F) unit **18**. The CPU **11**, the memory **12**, the storage unit **13**, the input unit **14**, the display unit **15**, the

medium reading and writing device **16**, and the communication I/F unit **18** are connected to each other via a bus **B1**. The medium reading and writing device **16** reads out information written in a recording medium **17** and writes information in the recording medium **17**.

(24) The storage unit **13** according to the present exemplary embodiment is realized by a hard disk drive (HDD), a solid state drive (SSD), a flash memory, or the like. An information processing program **13A** is stored in the storage unit **13** as a storage medium. The information processing program **13A** is stored (installed) in the storage unit **13** in a manner that the recording medium **17** in which the information processing program **13A** has been written is connected to the medium reading and writing device **16**, and the medium reading and writing device **16** reads out the information processing program **13A** from the recording medium **17**. The CPU **11** reads out the information processing program **13A** from the storage unit **13**, loads the information processing program **13A** into the memory **12**, and sequentially executes processes in the information processing program **13A**.

(25) A virtual printer management information database **13B** is stored in the storage unit **13**. The details of the virtual printer management information database **13B** will be described later.

(26) Next, the configuration of the terminal device **30** according to the present exemplary embodiment will be described with reference to FIG. **3**. FIG. **3** is a block diagram illustrating an example of the hardware configuration of the terminal device **30** according to the present exemplary embodiment.

(27) As illustrated in FIG. **3**, the terminal device **30** according to the present exemplary embodiment includes a CPU **31** as a processor, a memory **32** as a temporary storage area, a non-volatile storage unit **33**, an input unit **34** such as a keyboard and a mouse, a display unit **35** such as a liquid crystal display, a medium reading and writing (R/W) device **36**, and a communication I/F unit **38**. The CPU **31**, the memory **32**, the storage unit **33**, the input unit **34**, the display unit **35**, the medium reading and writing device **36**, and the communication I/F unit **38** are connected to each other via a bus **B2**. The medium reading and writing device **36** reads out information written in a recording medium **37** and writes information in the recording medium **37**.

(28) The storage unit **33** according to the present exemplary embodiment is realized by an HDD, an SSD, a flash memory, or the like. A printing instruction processing program **33A** is stored in the storage unit **33** as a storage medium. The printing instruction processing program **33A** is stored (installed) in the storage unit **33** in a manner that the recording medium **37** in which the printing instruction processing program **33A** has been written is connected to the medium reading and writing device **36**, and the medium reading and writing device **36** reads out the printing instruction processing program **33A** from the recording medium **37**. The CPU **31** reads out the printing instruction processing program **33A** from the storage unit **33**, loads the printing instruction processing program **33A** into the memory **32**, and sequentially executes processes in the printing instruction processing program **33A**.

(29) Next, the functional configuration of the server **10** according to the present exemplary embodiment will be described with reference to FIG. **4**. FIG. **4** is a block diagram illustrating an example of the functional configuration of the server **10** according to the present exemplary embodiment.

(30) As illustrated in FIG. **4**, the server **10** includes an execution unit **11A** and a reception unit **11B**. The CPU **11** in the server **10** executes the information processing program **13A**, and thereby the CPU **11** functions as the execution unit **11A** and the reception unit **11B**.

(31) In a case where the execution unit **11A** according to the present exemplary embodiment receives an instruction to apply a setting of a second image forming device **50** (referred to as a “second image forming device” below) to a first image forming device **50** (referred to as a “second image forming device” below), the execution unit **11A** executes an application process of applying a setting of a second virtual image forming device (referred to as a “second virtual image forming device” below) to a setting of a first virtual image forming device (referred to as a “first virtual

image forming device" below). The first virtual image forming device and the second virtual image forming device correspond to the above-described virtual printer.

(32) Here, the first virtual image forming device is a virtual device of the first image forming device, and is a device having a setting synchronized with the first image forming device. The second virtual image forming device is a virtual device of the second image forming device, and is a device having a setting synchronized with the second image forming device. Thus, with the application process of the execution unit **11A**, the setting of the second virtual image forming device applied to the setting of the first virtual image forming device, that is, the setting of the second image forming device is reflected to the first image forming device by the above synchronization.

(33) Further, the execution unit **11A** according to the present exemplary embodiment executes a saving process of saving the setting of the first virtual image forming device prior to the execution of the application process. After the first image forming device has ended an image process, the execution unit **11A** according to the present exemplary embodiment executes a return process of returning the setting of the first virtual image forming device to the setting saved by the above saving process.

(34) Here, in a case where the return process is executed, the execution unit **11A** according to the present exemplary embodiment excludes the setting related to the history of the image process executed by the first image forming device, from the target of the return process. This is to avoid erroneous return of the setting related to the history.

(35) Prior to the execution of the application process, the execution unit **11A** according to the present exemplary embodiment executes a verification process of verifying whether or not the setting of the second virtual image forming device can be applied to the setting of the first virtual image forming device. In a case where it is verified that the setting can be applied as a result of the verification process, the execution unit **11A** executes the application process.

(36) Here, the execution unit **11A** according to the present exemplary embodiment executes a static verification process and a dynamic verification process, as the verification process. In the static verification process, it is verified whether or not the setting of the second image forming device can be applied to the first image forming device. In the dynamic verification process, it is verified whether or not the setting of the second image forming device can be applied in the state of the first image forming device. The present disclosure is not limited to this form. A form in which only one of the static verification process and the dynamic verification process is executed as the above verification process may be made.

(37) Further, in a case where the execution unit **11A** according to the present exemplary embodiment verifies that the application is not possible as a result of the verification process, the execution unit **11A** executes a presentation process of presenting, to a user, information indicating that the application is not possible. In the present exemplary embodiment, a presentation by the display of the display unit is applied as the presentation by the presentation process, but the present disclosure is not limited to this. For example, a form in which a presentation by printing of the image forming device or a presentation by voice from a voice generation device is applied as the presentation by the presentation process may be made.

(38) Furthermore, in a case where it is verified that applying a portion of the setting is not possible as a result of the verification process, the execution unit **11A** according to the present exemplary embodiment selectively executes either a first process of applying only other portions of the setting or a second process of causing the application process not to be executed. Then, the reception unit **11B** according to the present exemplary embodiment receives the selection designation for which of the first process and the second process is to be executed. The execution unit **11A** executes the process designated by the selection designation received by the reception unit **11B**.

(39) Next, the virtual printer management information database **13B** according to the present exemplary embodiment will be described with reference to FIG. 5. FIG. 5 is a schematic diagram

illustrating an example of the configuration of the virtual printer management information database **13B** according to the present exemplary embodiment.

(40) The virtual printer management information database **13B** according to the present exemplary embodiment is a database in which information regarding the above-described virtual printer is registered. As illustrated in FIG. 5 as an example, information of a printer name and information of a setting are stored in association with each other.

(41) The printer name is information indicating the name of an image forming device **50** assigned in advance to the image forming device **50** in order to specify the image forming device **50** corresponding to the corresponding virtual printer. The setting is information (referred to as “setting information” below) set in the image forming device **50** corresponding to the corresponding virtual printer.

(42) As illustrated in FIG. 5, the setting information includes information (referred to as “default setting information” below) indicating settings to be applied in a case where image formation is performed, for example, “color mode” for setting whether to apply black-and-white printing or color printing, “two-sided printing” for setting whether to perform two-sided printing, and “stapler” for setting whether or not to stop a stapler. Further, the setting information also includes information (referred to as “history information” below) regarding the history of the image process executed previously by each image forming device **50**, for example, what process has been executed by each image forming device **50**.

(43) Next, an operation of the information processing system **1** according to the present exemplary embodiment will be described with reference to FIGS. 6 to 10.

(44) First, the operation of the terminal device **30** in a case of executing a printing instruction process will be described with reference to FIGS. 6 to 7. FIG. 6 is a flowchart illustrating an example of the printing instruction process according to the present exemplary embodiment.

(45) In the information processing system **1** according to the present exemplary embodiment, a user of the terminal device **30** (referred to as a “printing instructor” below) instructs to print a printed matter (referred to as a “target printed matter” below) set as a target, by using the terminal device **30**. In a case where the printing instructor issues a printing instruction, the CPU **31** in the terminal device **30** executes the printing instruction processing program **33A**. Thus, the printing instruction process illustrated in FIG. 6 is executed.

(46) In Step **S100** in FIG. 6, the CPU **31** controls the display unit **35** to display a printing instruction screen having a predetermined configuration. In Step **S102**, the CPU **31** waits until predetermined information is input. FIG. 7 illustrates an example of the printing instruction screen according to the present exemplary embodiment.

(47) As illustrated in FIG. 7, a message for urging the designation of an image forming device or the like that performs printing of the target printed matter is displayed on the printing instruction screen according to the present exemplary embodiment. Further, information indicating the name of the image forming device **50** that can be used in the terminal device **30** is displayed on the printing instruction screen according to the present exemplary embodiment. Further, a designation portion **35A** for designating an image forming device **50** to be used and an input area **35B** for inputting the name of an image forming device **50** having a setting that is intended to be applied to the image forming device **50** are displayed on the printing instruction screen according to the present exemplary embodiment.

(48) In a case where the printing instruction screen illustrated in FIG. 7 is displayed, the printing instructor uses the input unit **34** to designate the designation portion **35A** corresponding to the image forming device **50** that prints the target printed matter. In a case where the printing instructor intends to apply the setting of another image forming device **50** to the designated image forming device **50** by using the input unit **34**, the printing instructor inputs the name of the other image forming device **50** into the corresponding input area **35B**. In a case where the designation or the input of the image forming device **50** is ended, the printing instructor designates an end button **35C**

by using the input unit **34**. In a case where the printing instructor designates the end button **35C**, Step **S102** is determined to be positive. Thus, the process proceeds to Step **S104**.

(49) In Step **S104**, the CPU **31** creates printing job information for the target printed matter. At this time, in a case where the name of the image forming device **50** is input into the corresponding input area **35B** by the printing instructor, the CPU **31** creates printing job information including information (referred to as “target device information” below) indicating this image forming device **50** (corresponding to the above-described second image forming device, and referred to as the “second image forming device” below).

(50) The CPU **31** transmits the created printing job information to the image forming device **50** (corresponding to the above-described first image forming device, and referred to as the “first image forming device” below) designated in the designation portion **35A**. Then, the CPU **31** ends the present printing instruction process.

(51) In the present exemplary embodiment, as described above, the selection designation of the image forming device **50** to be used and the input of the image forming device **50** to be set are performed through the dedicated screen illustrated in FIG. **7** as an example. The present disclosure is not limited to this. For example, a form in which the above designation is performed by using a standard function provided in a browser may be made.

(52) Upon receiving the printing job information, the first image forming device temporarily stores the received printing job information. The first image forming device performs login authentication for the printing instructor, and then performs printing of a printed matter using the received printing job information, in response to an instruction by the printing instructor.

(53) Here, in a case where the printing job information as a printing target includes the target device information, the first image forming device according to the present exemplary embodiment transmits the target device information to the server **10** prior to printing of the printed matter.

(54) Next, the operation of the server **10** in a case of executing information processing will be described with reference to FIGS. **8** to **10**. FIG. **8** is a flowchart showing an example of information processing according to the present embodiment, and FIG. **9** is a flowchart showing an example of verification process according to the present exemplary embodiment.

(55) Regarding the information processing according to the present exemplary embodiment, the information processing illustrated in FIG. **8** is performed in a manner the CPU **11** in the server **10** executes the information processing program **13A** in a case where the server **10** receives target device information from a certain image forming device **50**.

(56) In Step **S200** in FIG. **8**, the CPU **11** executes the above-described saving process by reading out information (referred to as “first setting information” below) indicating the setting of the virtual printer (corresponding to the above-described first virtual image forming device, and referred to as the “first virtual image forming device” below) corresponding to the first image forming device that is a transmission source of the target device information, from the virtual printer management information database **13B**, and storing the read-out information in the storage unit **13** again. At this time, the CPU **11** stores only the default setting information among pieces of information indicating the setting, as the first setting information.

(57) In Step **S202**, the CPU **11** reads out information (referred to as “second setting information” below) indicating the setting of the virtual printer (corresponding to the above-described second virtual image forming device, and referred to as the “second virtual image forming device” below) corresponding to the second image forming device indicated by the received target device information, from the virtual printer management information database **13B**. At this time, the CPU **11** reads out only the default setting information among the pieces of information indicating the setting, as the second setting information.

(58) In Step **S204**, the CPU **11** executes the verification process illustrated in FIG. **9** as an example, by using the first setting information and the second setting information. In the verification process, it is verified whether or not the setting of the second virtual image forming device can be applied to

the setting of the first virtual image forming device

(59) In Step S250 in FIG. 9, the CPU 11 executes the static verification process described above, and determines whether or not the verification result by the static verification process has no problem. In a case where Step S250 is determined to be positive, the CPU 11 causes the process to proceed to Step S252. In the present exemplary embodiment, a process of determining whether or not setting items indicated by the first setting information include all setting items identical to setting items indicated by the second setting information is applied as the static verification process. The present disclosure is not limited to this. For example, a form in which a process of determining whether or not the setting items indicated by the first setting information include a setting item that can be substituted with a setting item although this setting item different from the setting items indicated by the first setting information is included in the setting items indicated by the second setting information is applied as the static verification process may be made.

(60) In Step S252, the CPU 11 executes the above-described dynamic verification process, and determines whether or not the verification result by the dynamic verification process has no problem. In a case where Step S252 is determined to be positive, the CPU 11 causes the process to proceed to Step S254. In the present exemplary embodiment, a process of determining whether or not all optional items targeted in the setting items indicated by the second setting information are mounted in the first image forming device is applied as the dynamic verification process, but the present disclosure is not limited to this. For example, a form in which a process of determining whether or not a substitute for optional items is attached to the first image forming device although some of optional items targeted by the setting items indicated by the second setting information are temporarily detached from the first image forming device is applied as the dynamic verification process may be made.

(61) In Step S254, the CPU 11 overwrites the first setting information in the virtual printer management information database 13B with the second setting information, thereby replacing the first setting information with the second setting information. Then, the CPU 11 ends the present verification process. By replacing the setting information, the above-described application process is executed.

(62) On the other hand, in a case where either Step S250 or Step S252 is determined to be negative, the process proceeds to Step S256. In Step S256, the CPU 11 displays a setting selection screen for the first image forming device by transmitting information that can display the setting selection screen having a predetermined configuration to the first image forming device. In Step S258, the CPU 11 waits until predetermined information is received from the first image forming device. FIG. 10 illustrates an example of the setting selection screen according to the present exemplary embodiment.

(63) As illustrated in FIG. 10, information indicating a problem found by the static verification process or the dynamic verification process is displayed on the setting selection screen according to the present exemplary embodiment. Further, information for urging selection designation of whether to apply the setting other than the setting corresponding to the displayed problem (corresponding to the first process, and referred to as the “first process” below) or not apply all the settings (corresponding to the second process, and referred to as the “second process” below) is displayed on the setting selection screen according to the present exemplary embodiment. Further, a first button 52A designated in a case where the first process is adopted and a second button 52B designated in a case where the second process is adopted are displayed on the setting selection screen according to the present exemplary embodiment.

(64) In a case where the setting selection screen illustrated in FIG. 10 is displayed, the printing instructor uses the input unit in the first image forming device to designate the first button 52A in a case where the first process is adopted, and designate the second button 52B in a case where the second process is adopted. In a case where the printing instructor designates the first button 52A or the second button 52B, the first image forming device transmits, to the server 10, information

(referred to as “adoption process information” below) indicating the process corresponding to the designated button. In a case where the adoption process information is received from the first image forming device, Step S258 is determined to be positive. Thus, the process proceeds to Step S260.

(65) In Step S260, the CPU 11 executes the process indicated by the received adoption process information, and then ends the present verification process. In a case where the verification process is ended, the process returns to Step S206 of the information processing.

(66) With the above processes, at least some of the settings of the first virtual image forming device are replaced with at least some of the settings of the second virtual image forming device, or the settings are not replaced.

(67) Thus, in a case where the above settings are replaced, the settings of the first image forming device are subsequently synchronized with the settings of the first virtual image forming device. Thus, at least some of the settings of the second image forming device are reflected to the first image forming device.

(68) Then, as described above, the printing instructor instructs the first image forming device to print the target printed matter. Thus, the first image forming device performs printing of the target printed matter. In a case where the printing of the target printed matter is ended, the first image forming device transmits, to the server 10, information indicating that the printing has been ended (referred to as “printing end information” below).

(69) Therefore, in Step S206, the CPU 11 waits until the print end information is received. In Step S208, the CPU 11 returns the first setting information to the original information by overwriting the information indicating the setting of the first virtual image forming device with the first setting information saved by the process in Step S200. Then, the CPU 11 ends the present information processing. By returning the setting information, the above-described return process is executed.

Second Exemplary Embodiment

(70) In the present exemplary embodiment, a form example in which a duplication creation process of creating a duplication of the first virtual image forming device is executed will be described. Since the overall configuration of the information processing system 1 to the present exemplary embodiment is substantially similar to the overall configuration in the first embodiment, here, first, the functional configuration of a server 10 according to the present exemplary embodiment will be described with reference to FIG. 11. FIG. 11 is a block diagram illustrating an example of the functional configuration of the server 10 according to the present exemplary embodiment. The identical components to the components illustrated in FIG. 4 are denoted by the identical reference signs to the reference signs in FIG. 4, and the description will not be repeated.

(71) As illustrated in FIG. 11, the server 10 according to the present exemplary embodiment is different from the server 10 according to the first exemplary embodiment only in that an execution unit 11C is applied instead of the execution unit 11A.

(72) That is, the execution unit 11C according to the present exemplary embodiment executes the duplication creation process of creating a duplication of the first virtual image forming device prior to the execution of the application process, in addition to the processes substantially similar to the processes of the execution unit 11A according to the first exemplary embodiment.

(73) Next, the operation of an information processing system 1 according to the present exemplary embodiment will be described. Since the operation of a terminal device 30 in a case of executing the printing instruction process is identical to the operation in the first exemplary embodiment, the description thereof will be omitted here. The operation of the server 10 in a case where information processing is performed will be described below with reference to FIGS. 12 to 13. FIG. 12 is a flowchart illustrating an example of the information processing according to the present exemplary embodiment, and FIG. 13 is a flowchart illustrating an example of a verification process according to the present exemplary embodiment. The steps in which the identical processing is performed illustrated in FIGS. 8 and 9 are denoted by the step numbers identical to the step numbers in FIGS.

8 and **9**, and the description thereof will not be repeated.

(74) As illustrated in FIG. **12**, the information processing according to the present exemplary embodiment is different from the information processing according to the first exemplary embodiment in that the process of Step **S201** is applied to the information processing according to the first exemplary embodiment instead of the process of Step **S200**, and the process of Step **S205** is applied instead of the process of Step **S204**, and the process of Step **S209** is applied instead of the process of Step **S208**.

(75) That is, in Step **S201** of the information processing according to the present exemplary embodiment, the CPU **11** executes the duplication creation process of creating a duplication of the first virtual image forming device instead of the saving process.

(76) That is, in the present exemplary embodiment, the duplication creation process is executed by storing the information indicating the first virtual image forming device itself (referred to as “first virtual image forming device information” below) in the storage unit **13** again.

(77) Further, in Step **S205**, the CPU **11** executes a verification process different from the verification process in the first exemplary embodiment, which is illustrated in FIG. **13** as an example. As illustrated in FIG. **13**, the verification process according to the present exemplary embodiment is different from the verification process in the first exemplary embodiment, in that the process of Step **S255** is applied instead of the process of Step **S254**, and the process of Step **S261** is applied instead of the process of Step **S260**.

(78) That is, in Step **S255** of the verification process according to the present exemplary embodiment, the CPU **11** replaces the first setting information in the first virtual image forming device information with the second setting information by overwriting the information indicating the setting of the first virtual image forming device information duplicated by the duplication creation process with the second setting information.

(79) Further, in Step **S261**, the CPU **11** executes the process indicated by the adoption process information received from the first image forming device, and then ends the verification process. In addition, a point that the process of Step **S261** according to the present exemplary embodiment targets the setting of the duplicated first virtual image forming device information is different from a point that the process of Step **S260** according to the first exemplary embodiment targets the setting of the first virtual image forming device itself.

(80) On the other hand, in Step **S209** of the information processing, the CPU **11** erases the duplication created by the process of Step **S201**, and then ends the present information processing.

(81) In a case where the duplicated first virtual image forming device information exists, the first image forming device according to the present exemplary embodiment synchronizes the setting with the first virtual image forming device information.

(82) With the above processing, it is possible to reflect the setting of the second image forming device to the first image forming device without changing the setting of the first virtual image forming device. Therefore, in the information processing system **1** according to the present exemplary embodiment, even in a case where the first image forming device is shared by a plurality of printing instructors, it is possible to reflect the setting of the second image forming device to the first image forming device without limiting the time of use.

(83) In each of the above exemplary embodiments, the case where the present invention is applied to the cloud print service has been described, but the present disclosure is not limited to this. For example, a form in which the present invention is applied to a print service that does not use a cloud server, for example, a network server in an organization may be made.

(84) Further, in each of the above exemplary embodiments, the case where the virtual printer management information database **13B** is registered in the server **10** has been described, but the present disclosure is not limited to this. For example, a form in which the virtual printer management information database **13B** is registered in any image forming device **50** or another device accessible from any image forming device **50** may be made.

(85) Further, in each of the above exemplary embodiments, the case where the image forming device is applied as the image processing device in the present disclosure has been described, but the present disclosure is not limited thereto. For example, a form in which another image processing device such as an image scanning device having only an image scanning function or an image transmission device having an image transmission function in addition to the image scanning function is applied as the image processing device of the present disclosure may be made.

(86) Although the exemplary embodiments have been described above, the technical scope of the present invention is not limited to the scope described in the above exemplary embodiments. Various changes or improvements may be made to the above exemplary embodiments without departing from the gist of the invention, and the changed or improved forms are also included in the technical scope of the invention.

(87) Further, each of the above exemplary embodiments does not limit the invention according to the claim, and not all combinations of characteristics described in the exemplary embodiments are not limited to being required for the means for addressing the invention. The above-described exemplary embodiments include inventions at various stages, and various inventions are extracted by combining a plurality of disclosed constituent requirements. Even though some constituent requirements are deleted from all the constituent requirements described in the exemplary embodiments, a configuration in which some of the constituent requirements are deleted may be extracted as an invention as long as the effect is obtained.

(88) In the embodiments above, the term “processor” refers to hardware in a broad sense. Examples of the processor include general processors (e.g., CPU: Central Processing Unit) and dedicated processors (e.g., GPU: Graphics Processing Unit, ASIC: Application Specific Integrated Circuit, FPGA: Field Programmable Gate Array, and programmable logic device). In the embodiments above, the term “processor” is broad enough to encompass s one processor or plural processors in collaboration which are located physically apart from each other but may work cooperatively. The order of operations of the processor is not limited to one described in the embodiments above, and may be changed.

(89) Further, in each of the above exemplary embodiments, the case where various processes are realized by a software configuration using a computer by executing a program has been described, but the present invention is not limited to this. For example, a form in which various processes are realized by a combination of a hardware configuration and a software configuration may be made.

(90) In addition, the configurations of the server **10** and the terminal device **30** described in each of the above exemplary embodiments are examples. A unnecessary portion may be deleted or a new portion may be added, within a range without departing from the gist of the present invention.

(91) Further, the flow of various processes described in each of the above exemplary embodiments is also an example. An unnecessary step may be deleted, a new step may be added, or the processing order may be changed, within a range without departing from the gist of the present invention.

(92) The foregoing description of the exemplary embodiments of the present invention has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Obviously, many modifications and variations will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, thereby enabling others skilled in the art to understand the invention for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the following claims and their equivalents.

Claims

1. An information processing apparatus comprising: a processor configured to: based on a determination to apply a setting of a second image processing device to a setting of a first image processing device, execute an application process of applying a setting of a second virtual image processing device to a setting of a first virtual image processing device, the first virtual image processing device being a virtual device of the first image processing device and being a device having the setting synchronized with the setting of the first image processing device, and the second virtual image processing device being a virtual device of the second image processing device and being a device having the setting synchronized with the setting of the second image processing device.
2. The information processing apparatus according to claim 1, wherein the processor is configured to: prior to the execution of the application process, execute a saving process of saving the setting of the first virtual image processing device.
3. The information processing apparatus according to claim 2, wherein the processor is configured to: after an image process by the first image processing device has been ended, execute a return process of returning the setting of the first virtual image processing device to the setting saved by the saving process.
4. The information processing apparatus according to claim 3, wherein the processor is configured to: based on a determination that the return process is executed, exclude a setting related to a history of the image process executed by the first image processing device, from a target of the return process.
5. The information processing apparatus according to claim 4, wherein the processor is configured to: prior to the execution of the application process, execute a duplication creation process of creating a duplication of the first virtual image processing device.
6. The information processing apparatus according to claim 4, wherein the processor is configured to: prior to the execution of the application process, execute a verification process of verifying whether or not the setting of the second virtual image processing device is applicable to the setting of the first virtual image processing device, and based on a determination that the application is possible as a result of the verification process, execute the application process.
7. The information processing apparatus according to claim 3, wherein the processor is configured to: prior to the execution of the application process, execute a duplication creation process of creating a duplication of the first virtual image processing device.
8. The information processing apparatus according to claim 3, wherein the processor is configured to: prior to the execution of the application process, execute a verification process of verifying whether or not the setting of the second virtual image processing device is applicable to the setting of the first virtual image processing device, and based on a determination that the application is possible as a result of the verification process, execute the application process.
9. The information processing apparatus according to claim 2, wherein the processor is configured to: prior to the execution of the application process, execute a duplication creation process of creating a duplication of the first virtual image processing device.
10. The information processing apparatus according to claim 9, wherein the processor is configured to: prior to the execution of the application process, execute a verification process of verifying whether or not the setting of the second virtual image processing device is applicable to the setting of the first virtual image processing device, and based on a determination that the application is possible as a result of the verification process, execute the application process.
11. The information processing apparatus according to claim 2, wherein the processor is configured to: prior to the execution of the application process, execute a verification process of verifying whether or not the setting of the second virtual image processing device is applicable to the setting of the first virtual image processing device, and based on a determination that the application is possible as a result of the verification process, execute the application process.

12. The information processing apparatus according to claim 1, wherein the processor is configured to: prior to the execution of the application process, execute a duplication creation process of creating a duplication of the first virtual image processing device.
13. The information processing apparatus according to claim 12, wherein the processor is configured to: prior to the execution of the application process, execute a verification process of verifying whether or not the setting of the second virtual image processing device is applicable to the setting of the first virtual image processing device, and based on a determination that the application is possible as a result of the verification process, execute the application process.
14. The information processing apparatus according to claim 1, wherein the processor is configured to: prior to the execution of the application process, execute a verification process of verifying whether or not the setting of the second virtual image processing device is applicable to the setting of the first virtual image processing device, and based on a determination that the application is possible as a result of the verification process, execute the application process.
15. The information processing apparatus according to claim 14, wherein the processor is configured to: execute, as the verification process, at least one of a static verification process of verifying whether or not the setting of the second virtual image processing device is allowed to be applied to the first virtual image processing device, or a dynamic verification process of verifying whether or not the setting of the second virtual image processing device is allowed to be applied in a state of the first virtual image processing device.
16. The information processing apparatus according to claim 14, wherein the processor is configured to: based on a determination that the application is not possible as a result of the verification process, execute a presentation process of presenting, to a user, information indicating that the application is not possible.
17. The information processing apparatus according to claim 14, wherein the processor is configured to: based on a determination that applying a portion of the setting is not possible as a result of the verification process, selectively execute either a first process of applying only other portions of the setting or a second process of causing the application process not to be executed.
18. The information processing apparatus according to claim 17, wherein the processor is configured to: receive selection designation for which of the first process and the second process is to be executed, and execute a process designated by the received selection designation.
19. A non-transitory computer readable medium storing an information processing program causing a computer to execute a process comprising: executing, based on a determination to apply a setting of a second image processing device to a setting of a first image processing device, an application process of applying a setting of a second virtual image processing device to a setting of a first virtual image processing device, the first virtual image processing device being a virtual device of the first image processing device and being a device having the setting synchronized with the setting of the first image processing device, and the second virtual image processing device being a virtual device of the second image processing device and being a device having the setting synchronized with the setting of the second image processing device.
20. An information processing method comprising: executing, based on a determination to apply a setting of a second image processing device to a setting of a first image processing device is received, an application process of applying a setting of a second virtual image processing device to a setting of a first virtual image processing device, the first virtual image processing device being a virtual device of the first image processing device and being a device having the setting synchronized with the setting of the first image processing device, and the second virtual image processing device being a virtual device of the second image processing device and being a device having the setting synchronized with the setting of the second image processing device.
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